<table>
<tr><th colspan="4" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="6">Marks Available</th></tr>
<tr><th>Total</th><th>AO1</th><th>AO2</th><th>AO3</th><th>Maths</th><th>Prac</th></tr>
<tr><td>11.</td><td>(a)</td><td></td><td></td><td>award (1) for either of following
a molecule that partially dissociates to release/donate H^+
a molecule that releases H^+ in a reversible reaction</td><td>1</td><td>1</td><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td>(b)</td><td></td><td></td><td>award (1) each for calculating K_a or pK_a or pH for two acids
<table>
<tr><th></th><th>K_a / mol dm^{-3}</th><th>pK_a</th><th>pH</th></tr>
<tr><td>X = H</td><td>1.32×10^{-5}</td><td>4.88</td><td>2.59</td></tr>
<tr><td>X = Cl</td><td>1.48×10^{-3}</td><td>2.83</td><td>1.57</td></tr>
<tr><td>X = OH</td><td>1.38×10^{-4}</td><td>3.86</td><td>2.08</td></tr>
</table>
X = H < X = OH < X = Cl (1)

ecf possible from calculated values</td><td>3</td><td></td><td>1</td><td>1

1</td><td>2</td><td></td></tr>
<tr><td></td><td>(c)</td><td>(i)</td><td></td><td>$[H^+] = \frac{K_w}{[OH^-]} = 1.25 \times 10^{-14}$ mol dm^{-3} (1)

pH = 13.9 (1)

ecf possible from incorrect $[H^+]$</td><td>2</td><td></td><td>2</td><td></td><td>2</td><td></td></tr>
<tr><td></td><td></td><td>(ii)</td><td></td><td>concentration of salt = concentration of acid so pH = pK_a (1)

pH = 4.74 (1)</td><td>2</td><td></td><td>1</td><td>1</td><td>2</td><td></td></tr>
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